

Summary

- Four technology transformations, each at the significant inflection point where the market completely redefined the business: eBay/PayPal from commerce into digital payments, Riverdeep into Houghton Mifflin Harcourt from traditional publishing into digital learning, Adobe from desktop software through cloud, data, and AI, and Silver Labs from a consumer product into an enterprise AI platform. Started as a founder, four products, two acquired, before spending over a decade building platforms inside Adobe.
- Architected and led the engineering build of Adobe's global product data and intelligence platform across **1500+ applications, 500M+ users, 140B+ daily interactions**, enabling **4.5× revenue growth (\$4B to \$19B+)**. Inside that platform, led both the architecture and the engineering delivery that brought Adobe Firefly, Adobe Express, Photoshop, and the Content Authenticity Initiative under a unified set of data standards and a common provenance model, including CAI/C2PA content provenance tagging that disclosed AI-generated media to millions of users, years before AI disclosure became a regulatory expectation.
- Carried that same discipline into an enterprise generative AI platform, designing and building deterministic control layers over probabilistic AI systems producing **200K+ videos** across **80+ integrations**, cutting output inconsistency by **97%**, production time by **~12×**, while holding **99.999% platform reliability**.
- Hands-on with a bias toward action. Built lean, globally distributed engineering teams that ship production systems at enterprise scale. Deep experience across privacy, compliance, and trust, GDPR, CCPA, COPPA, EU AI Act, DSA, working directly with legal, privacy, and trust & safety on regulatory design, not after the fact.

Experience
Adobe, Silver Labs 2012– Present

Chief Architect, CTO, EVP of Engineering, Platform & Product, Silver Labs 2024 – Present

- Architected and engineered a data-driven generative AI system, content ingestion, LLM-based generation, multimodal intelligence, voice synthesis, automated distribution, generating **400K+ videos** across **80+ integrations** and **15 distribution channels**, reaching **1B+ viewers** at **99.999% platform reliability**.
- Designed and engineered a deterministic control architecture for probabilistic AI systems, staged pipelines, validation layers, model routing, cutting output inconsistency by **97%**, production time from **~4h to ~20m (~12×**), regeneration rates by **50%**, and manual intervention by **75%**. Applied the same governance discipline pioneered through the CAI/C2PA work directly into the generation pipeline, rather than as validation after generation.
- Designed and built LLM-driven generation systems enforcing **20+ semantic and editorial constraints**, producing high-fidelity, auditable output through constraint-driven inference.
- Developed multimodal intelligence systems for signal extraction across text, audio, and visual inputs, improving scene selection accuracy by **57%** within the automated generation pipeline.
- Re-architected voice synthesis pipelines with orchestration, normalization, and optimization layers, cutting defects **~50%** and generation cost **~25%**.
- Architected and developed Engineering OS, an AI-assisted engineering platform codifying **20+ reusable engineering skills** spanning architecture, data, distributed systems, and deterministic AI into reusable workflows, enabling engineers and AI agents to consistently deliver production-quality systems.
- Built and led a lean, globally distributed data, AI, and product engineering org combining human engineers and AI agents to deliver production-scale generative AI systems while remaining hands-on across architecture, engineering, and delivery.

Principal Scientist & Head, Data, Analytics & Intelligent Platforms, Adobe (Architecture & Engineering | SVP org) 2012 – 2024

- Defined the architecture and led the engineering build of company-wide data, intelligence, and platform systems across **1500+ applications, 500M+ users**, and **140B+ daily interactions**, enabling Adobe's growth from **\$4B to \$19B+ revenue**. Designed, implemented, and operated a **90PB+ data lake** processing **300 trillion objects** and ingesting **100TB+ daily**, providing the data foundation for analytics, experimentation, AI, and product intelligence across business units and product ecosystem. Presented at the **AWS re:Invent** keynote on real-time data with the GM and VP of AWS Serverless.
- Led both the architecture and the engineering delivery that brought Adobe Firefly, Adobe Express, the Content Authenticity Initiative, and Photoshop under one shared architecture and standard data model, one bar across generative AI and creative products.
- Designed and delivered CAI/C2PA content provenance integration with Firefly, tagging AI-generated images at the point of creation to establish provenance and transparency for **~2M users**, shipped with the Firefly Style Match feature at Adobe MAX. One of the earliest production deployments of AI-generation disclosure at consumer scale, years ahead of current AI governance regulation.
- Designed and built the direct feedback loop from Firefly generation telemetry into RLHF-based model retraining, including detection of harmful prompts, bad actors, and account abuse, creating a closed-loop feedback system between generation quality, trust signals, and model improvement.
- Architected and engineered privacy-by-design data and AI systems, data minimization, lineage, policy-driven access control (POLP), on-device encryption, in-motion masking, enabling GDPR and CCPA-compliant real-time decisioning across **~2B devices**.
- Partnered directly with Legal, Privacy, Product, Security, and executive leadership to translate emerging regulatory requirements, including the DSA, into production platform capabilities.
- Established canonical data models (**120+ attributes**) and extensible standards across **110+ teams**, enabling analytics, experimentation, AI/LLM systems, and new product capabilities without re-architecture.
- Implemented observability and reliability systems across **2K+ dashboards**, cutting detection and response time **~95%** for **~10K engineers and scientists** depending on the platform daily.
- Owned and optimized **~\$20M** in infrastructure spend, reducing cost by **~\$5M** without loss of reliability.
- Built and led a globally distributed platform engineering org enabling **~30K engineers, product managers, and designers** across Adobe's product ecosystem, without direct authority over most of them. Delivered **\$150M+** incremental impact and **\$110M ARR** through platform-enabled experimentation and analytics.

Prior Experience | Founding member, CTO, Chief Architect, Acquisitions & Exits Pre-2012

- eBay / PayPal, architected and built shared platform systems generating **~\$600/sec** in transaction revenue and serving **~40K businesses** as commerce expanded into marketplace and digital payments.
- Riverdeep to Houghton Mifflin Harcourt, designed and built digital learning and assessment platforms for **~400K students** and **~20K teachers**, contributed to a patent on blended learning systems.
- Judicial Council of California, architected and delivered government transaction and analytics systems for **~30M citizens**, delivering **~\$60M** in savings on a **\$540M annual budget**.
- Delivered early cloud and SaaS platforms, including a **~\$10M ARR** architecture initiative for Fortune 500 customers and a **~\$1M** cloud orchestration platform, alongside airline reservation, operations, and tax & compliance systems.

Platforms, Data & AI Systems

- Data Platforms & Cloud:** AWS (Kinesis, Lambda, S3, EMR), Databricks, large-scale data and AI platform infrastructure.
- Streaming & Distributed Systems:** Kafka, Spark, Flink, real-time analytics and AI-driven systems.
- AI & Generative Systems:** LLM pipelines, prompt and instruction engineering, frontier models (OpenAI, Claude), orchestration frameworks (LangChain), vector search and knowledge platforms (MongoDB), multimodal systems, content provenance and authenticity, voice synthesis, production-scale generative AI platforms.
- Infrastructure & Platform Systems:** Serverless architectures, Kubernetes, observability and reliability systems, experimentation platforms.
- Trust, Governance & Compliance:** GDPR, CCPA, COPPA, EU AI Act, C2PA / Content Authenticity Initiative, Privacy & Governance by Design, Digital Services Act (DSA), Principle of Least Privilege (POLP), AI content provenance and disclosure, RLHF trust and safety feedback loops, U.S. federal and enterprise compliance.

Education
Bachelor of Engineering, Computer Science, University of Bombay, First Class with Honors.

Executive snapshot

- Products and platforms reaching **1B+ users**, with repeated **0-to-1 and 1-to-100** launches
- Systems supporting **1.2M requests/second** at **~6 ms**, with **99.9999% reliability**
- Data & Intelligence platforms processing **100 TB+ daily** across **330+ pipelines**, a **90 PB+ data lake** and **300T+ objects**
- Built and led distributed teams across the **US, India and Europe**
- Helped lead Adobe’s cloud, data and AI transformation during its growth from **\$4B to \$19B+**
- Contributed to launches including **Firefly, Adobe Express, Photoshop Web, Content Authenticity and Experience Cloud**
- Startup experience spanning **two exits** and advisory work

AI & Innovation

- Author, **AI Instruction Design Set (AI-IDS) Architecture**, for high-determinism, low-variance production AI systems; under moderation for publication on **arXiv**, where Cornell University is a founding institutional member
- Creator, **Engineering OS**, integrating AI skills, frameworks and workflows for high-confidence engineering execution; implemented across three companies, including two Fortune 500 enterprises, and being finalized for open source
- Inventor, **Product Engineering Execution Framework**; patent pending
- Patent contributions spanning **AI feedback, learning systems, data models, analytics, reliability, cloud optimization and product execution**
- Recognized by AWS for pioneering **serverless computing at enterprise scale**
- Highest audience vote at an **AWS re:Invent keynote session** on real-time data
- Adobe Founders’ Award nominee** for innovation and business, technology and people transformation, Prior awards at eBay, CODie, Houghton Mifflin

Leadership Breadth

- Product and Platform Engineering, AI & GenAI, Cloud and Data Systems, APIs and Marketplaces
- Trust and Provenance, Privacy and Compliance
- B2B, B2C, B2B2C, Startups, Global Enterprises

Teaching & Impact

- Adjunct Professor, **AI & Emerging Technologies**, San José State University
- Leading **technology strategy and implementation** for CWC’s cancer initiative
- Guest speaker, **AWS Serverless Annual All-Hands**
- Advisor to **Adobe and Green Ninja Science**
- Helping mechanics, teachers and nonprofits build their online presence